

Master Thesis



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F3

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Neurogenesis of Neural Networks for 3d Nesting

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Field of study: Artificial Intelligence
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Acknowledgements

Děkuji ČVUT, že mi je tak dobrou *alma mater*.

Declaration

Prohlašuji, že jsem předloženou práci vypracoval samostatně, a že jsem uvedl veškerou použitou literaturu.

V Praze, 22. August 2025

Abstract

This thesis proposes a learning-guided system for 3D nesting and transport of convex items that optimizes utilization under non-overlap, transport and container-bound constraints. It aims to explore use of neurogenesis in order to appraise its viability for this class of problems.

It proposes state exploration using heuristics to enumerate feasible loading plans combined with neural scoring models and architectures to rank candidate solutions. Model architectures are to be evolved via neurogenesis.

Items, containers, and candidates are represented with compact local and global features capturing size, capacity usage, and transport attributes. We survey the state of the art and build a reproducible benchmark. Evaluation reports utilization, bin count, cost, and runtime.

Keywords: nesting, neurogenesis, artificial intelligence, bin packing

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Abstrakt

Tato práce navrhuje systém pro 3D nesting a transport konvexních objektů řízený učením, který optimalizuje využití prostoru při dodržení vymezení jako je nepřekrývání, přepravních požadavků a atributy kontejneru. Cílem je prozkoumat použití neurogeneze a posoudit její vhodnost pro tuto třídu úloh.

Navrhuje prozkoumávání stavového prostoru pomocí heuristik, které enumerují proveditelné plány nakládky, v kombinaci s neuronovými skórovacími modely a architekturami pro řazení kandidátních řešení. Architektury modelů jsou vyvíjeny evolučně pomocí neurogeneze.

Položky, kontejnery a kandidáti jsou reprezentováni kompaktními lokálními i globálními vlastnostmi zachycujícími rozměry, využití kapacity a přepravní atributy. Posuzuje současné state-of-the-art řešení této třídy problému a navrhuje reprodukovatelný benchmark. Vyhodnocení bude uvádět míru využití, počet binů, náklady a dobu běhu.

Klíčová slova: polohování, neurogeneze, umělá inteligence, bin packing

Překlad názvu: Neurogeneze neuronových sítí pro 3D nesting

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Chapter 1

Introduction

The bin-packing problem is a common optimization challenge broadly that seeks to distribute items in a set number of bins while maximizing utilization. This thesis aims to address this problem in 3D environment, with additional constraints based upon shipping industry where this problem commonly occurs.

Classical heuristic methods are fast and effective in specific regimes, but their decision rules are difficult to tune and often fail to transfer when constraint sets or item distributions change. The core challenge is not generating feasible candidates that can be done deterministically, but choosing among many feasible placements under heterogeneous constraints.

This thesis argues that neurogenesis - evolutionary design of neural networks, could fit this setting. Neuroevolution can search over architectures and hyperparameters of compact neural scoring models that rank items, orientations, and placements, without requiring gradients through the discrete packing process. It accommodates black-box, multi-objective fitness and adapts naturally as constraint sets vary. We therefore combine a deterministic candidate generator with evolved scoring models, aiming for a practical solver that remains robust across shipping-specific constraints.



Chapter 2

Problem analysis

Packing and routing are tightly connected: assigning a set of customers to a vehicle determines a joint 3D load that must satisfy feasibility (in-bounds, non-overlap) and practical constraints such as stability, stacking, accessibility, and sequencing. In turn, what can be feasibly packed materially changes which tours are admissible and how long they are. Recent integrated models for vehicle routing with three-dimensional loading (3L-VRP) formalize and empirically confirm this interaction, showing that routing costs and optimal tours shift once realistic loading constraints are enforced.[AMSAVP22]

We therefore focus on a load-aware TSP: for a single vehicle, minimize tour length over its assigned customers while using a 3D packer as a feasibility oracle for the union of their items. We employ a NN supported packer that co-optimizes packing and routing.

For each candidate decision (which car to load, item orientation, and placement), the network not only scores geometric nesting efficiency but also considers final routing impact. This steers the system to co-locate geographically close deliveries in the same vehicle, shaping shorter routes while keeping loads feasible.[GPAAVP23]

Because this pipeline is discrete and multi-objective, we adopt neurogenesis to evolve the scoring network directly on outcome-based fitness of the TSP, avoiding gradients and adapting quickly and robustly as constraint sets vary. The following chapter formalizes the problem and specifies the subset of constraints addressed in this work.

2.1 Problem definition

2.1.1 Bin packing

Cutting and packing problems can be broadly summarized with following description.

Given are two sets of elements, namely

- a set of large objects (input, supply) and
- a set of small items (output, demand),

which which are defined in some number of geometric dimensions. Select some or all small items, group them into one or more subsets and assign each of the resulting subsets to one of the large objects such that the geometric condition holds, i.e. the small items of each subset have to be laid out on the corresponding large object such that

- all small items of the subset lie entirely within the large object and
- the small items do not overlap,

and a given (single-dimensional or multi-dimensional) objective function is optimised.[WHS07]

Definition 2.1 (3D bin packing). Let $I = \{1, \dots, n\}$ be items with sizes $(w_i, h_i, d_i) \in \mathbb{R}_{>0}^3$ and per-item sets of allowed orthogonal rotations $R_i \subseteq S_3$. Bins are identical axis-aligned boxes of size $(W, H, D) \in \mathbb{R}_{>0}^3$.

A *packing* assigns each item i a bin index $k \in \{1, \dots, K\}$, an orientation $r_i \in R_i$, and a position $\mathbf{x}_i = (x_i, y_i, z_i) \in \mathbb{R}_{\geq 0}^3$ such that:

$$(\text{in-bounds}) \quad 0 \leq x_i \leq W - w_i^{r_i}, \quad 0 \leq y_i \leq H - h_i^{r_i}, \quad 0 \leq z_i \leq D - d_i^{r_i},$$

$$(\text{non-overlap}) \quad \forall i \neq j \text{ in the same bin } k : (x_i + w_i^{r_i} \leq x_j) \vee (x_j + w_j^{r_j} \leq x_i) \vee \\ (y_i + h_i^{r_i} \leq y_j) \vee (y_j + h_j^{r_j} \leq y_i) \vee (z_i + d_i^{r_i} \leq z_j) \vee (z_j + d_j^{r_j} \leq z_i),$$

where $(w_i^{r_i}, h_i^{r_i}, d_i^{r_i})$ denotes the dimensions of item i permuted by r_i .

Objectives. Either (i) *min-bins*: pack all items using the minimum number K of bins; or (ii) *max-utilization*: given a bin budget K , pack a subset $J \subseteq I$ to maximize $\sum_{i \in J} w_i^{r_i} h_i^{r_i} d_i^{r_i}$ (equivalently, utilization per bin).

Additional constraints. Let $\mathcal{C}$ denote application-specific constraints (e.g., weight limits, stability/center-of-mass, accessibility/LIFO, orientation allowances, compatibility/segregation). A packing is $\mathcal{C}$ -feasible if it satisfies the geometric conditions above and all $c \in \mathcal{C}$.

We target 3D bin packing with multiple bins: given a set of rectangular boxes (convex items) with orthogonal rotations, pack them efficiently into minimum number of identical containers[WHS07] However, while optimal nesting of boxes is likely byproduct, it's overall secondary to the main objective function, as described later in the Traveling salesman problem section.2.1.3

2.1.2 Constraints

Cutting and packing problems are further defined by their constraints. Beyond the basic geometric feasibility (in-bounds, non-overlap), the literature further mentions

Orientation item-specific allowed orientations, fragile handling, stackability classes[dNAJ21]

Global capacity / weight per-bin weight limits, sometimes combined with volumetric limits[dNAJ21]

Stacking / load-bearing limits on vertical support to avoid damaging items, may depend on item type or allowable stack height.[GTMP22]

Stability & center of mass static equilibrium, and even dynamic stability formulations to avoid tip-over during handling or motion[GOGL16]

Accessibility LIFO or door-side accessibility so earlier-stop items are not blocked. Some works use soft penalization rather than hard constraints[BTC23]

Routing-coupled objectives the route's distance or time is optimized subject to feasible loadings per vehicle. Loading constraints materially change which tours are admissible. [KEK21]

2.1.3 Traveling salesman problem

We complement 3D bin packing with a routing objective so that packed vehicles travel as little as possible.

Definition 2.2 (Traveling Salesman problem). Let $G = (V, A)$ be a graph where V is a set of n vertices. A is a set of arcs or edges, and let $C : (C_{ij})$ be a distance (or cost) matrix associated with A . The TSP consists of determining a minimum distance circuit passing through each vertex once and only once[BTC23]

This work focuses on this problem as a

Objective.

$$\min_{\{a_{j,k}\}, \{\pi_k\}} \sum_{k=1}^K \left(d_{0, \pi_k(1)} + \sum_{i=1}^{m_k-1} d_{\pi_k(i), \pi_k(i+1)} + d_{\pi_k(m_k), 0} \right). \quad (2.1)$$

Subject to.

$$\sum_{k=1}^K a_{j,k} = 1, \quad \forall j \in V \quad (\text{coverage}) \quad (2.2)$$

$$\pi_k \text{ is a Hamiltonian tour on } S_k := \{j \in V \mid a_{j,k} = 1\}, \quad \forall k \quad (\text{tour feasibility}) \quad (2.3)$$

$$\text{PACK} \left(\bigcup_{j \in S_k} D_j \right) = \text{true}, \quad \forall k \quad (\text{loading feasibility}) \quad (2.4)$$

Notation.

- $V = \{1, \dots, n\}$: set of customers; 0 is the depot.
- K : number of vehicles (bins). $S_k = \{j \in V \mid a_{j,k} = 1\}$ are customers served by vehicle k ; $m_k = |S_k|$.
- $d_{i,j}$: travel distance between locations i and j .

- $a_{j,k} \in \{0, 1\}$: assignment of customer j to vehicle k .
- π_k : order (permutation) in which vehicle k visits all $j \in S_k$ (a Hamiltonian tour).
- D_j : set of packages (boxes) for customer j .
- $\text{PACK}(\cdot)$: 3D packing subroutine that enforces in-bounds, non-overlap, and any other shipping constraints.

The goal is to minimize the distance traveled by by reducing the number of containers, utilizing space efficiently and assigning objects with close destinations to the same car.

2.2 Neurogenesis

Neurogenesis or neuroevolution is a practice of evolving the architecture and hyperparameters of given neural scoring models with a genetic algorithm. Classical neurogenesis like NEAT established that evolving topology alongside weights is effective.[LSX⁺20] Latter techniques such as Evolutionary neural network search, shortly ENAS, focuses mostly on development of deep neural networks.[RMS⁺17]

Artificial neural network (NN) architecture design is a nontrivial and time-consuming task that often requires a high level of human expertise. Neurogenesis serves to automate the design of NN architectures and has proven to be successful in automatically finding NN architectures that outperform those manually designed by human experts.[BB24]

2.2.1 Principles

In principle, neural network design is viewed as a population-based search over both structure (topology/connection patterns) and parameters. Individuals encode candidate architectures and hyperparameters. Each is initiated, evaluated by an outcome-based fitness function. The target fitness is to be improved via actions of genetic algorithm of selection, crossover, mutation, and elitism.

2.2.2 Search space and genome encoding

We encode a candidate network as a vector of genes. Some examples include following, but compact search space keeps evaluation times manageable.[WSS⁺23]

- **Depth/width:** number of layers (e.g., 2–6); hidden sizes (e.g., 64/128/256/512).
- **Activations & normalization:** {ReLU, GELU, SiLU} and {none, LayerNorm, BatchNorm}.
- **Skip/residuals:** on/off per stage.
- **Context aggregation (optional):** {none, local MLP pooling, small GNN over kNN}.
- **Feature subset:** bitmask/index selecting which engineered features are used.
- **Regularization/training:** dropout rate, weight decay, learning-rate schedule.

Gene	Example domain
Depth	{2, 3, 4, 5, 6}
Hidden width	{64, 128, 256, 512}
Activation	{ReLU, GELU, SiLU}
Normalization	{none, LayerNorm, BatchNorm}
Residuals	{off, on}
Context aggregation	{none, MLP pooling, small GNN}
Feature subset	index / bitmask over features
Dropout	[0, 0.3]
Weight decay	[0, 10^{-3}]
LR schedule	{cosine, step, warmup}

Table 2.1: Genome (architecture & training) encoding for the scoring model.

2.2.3 Fitness design

Fitness is computed by outcome: each individual (architecture + hyperparameters) is instantiated, run on a batch of task instances, and mapped to a scalar for elitism. A simple scalarization is

$$\text{Fit} = -\text{TaskCost} - \lambda_{\text{viol}} \cdot \text{Violations} + \lambda_Q \cdot \text{Quality} - \lambda_t \cdot \text{Runtime},$$

with weights set to reflect priorities. Alternatively, multiple objectives can be kept in a Pareto set (non-dominated sorting).[EMH19] To reduce variance and cost, average fitness over small instance batches and use multi-fidelity evaluation (cheap proxies early; full fidelity for shortlisted elites).[WSS⁺23]

■ 2.2.4 Evolutionary operators

Evolutionary operators in neurogenesis follow standard genetic algorithm, specialized for neural networks by using a genome that encodes architecture and hyperparameters. The operators below are adapted to these NN-specific genomes.

Initialization. Sample genomes uniformly; optionally seed a few hand-designed baselines.

Selection. Tournament or rank-based selection; for distance vs. runtime trade-offs, use non-dominated sorting (NSGA-II style).

Crossover. One-point or uniform crossover for discrete genes; arithmetic/BLX- α blending for continuous ones.[TK01]

Mutation. Small edits - flip activation/normalization, ± 1 layer, resample feature subset, jitter dropout/weight decay, optionally self-adaptive mutation rates.[LSX⁺20]

Elitism, replacement. Keep the top E elites; fill the rest with best offspring; optionally add age/diversity pressure.



Chapter 3

State-of-the-art solutions

After defining the problem of three-dimensional nesting and motivating the need for an efficient, adaptive optimization solution, this chapter reviews the current state-of-the-art in related methods. The goal is to summarize the existing approaches and highlight the research gaps that motivate the proposed neurogenetic framework.

The chapter is divided into three main sections. The first section introduces nesting, focusing on heuristic and metaheuristic methods developed for the 3D bin packing and cutting stock problems. The second section explores evolutionary neural architecture search and related neurogenesis methods of automated neural network design. Finally, the third section revisits the traveling salesman problem (TSP) as it is used as part of a benchmark for testing generalization and efficiency of optimization algorithms in this work.



3.1 Nesting

This section categorizes solution strategies for the class of nesting problems. Individual proposed approaches can be broadly split by several characteristics which play a crucial role in determining how algorithm may approach nesting. They provide a handy overview when evaluating an approach's usefulness in particular scenarios, so I shall list several of them at the beginning of this section.

Rest of the section outlines current state-of-the-art algorithms for class of nesting based problems.

Information model (offline vs. online). In the offline setting, the entire item set is known, enabling pre-sorting and deeper look-ahead; online packing processes items as they arrive and must make irrevocable decisions. State-of-the-art strategies in both regimes share a constructive skeleton (candidate generation $\rightarrow$ selection) but differ in how they balance exploration, robustness, and speed; online variants typically adopt myopic but well-regularized scoring rules and occasional re-packing windows [ARCO22].

Space representation (rasterization vs. geometry-first). Raster/voxel models offer simple, parallelizable collision checks and convenient stability features at the cost of resolution dependence. Geometry-first models rely on exact or approximate intersection tests for convex bodies, oriented bounding boxes, separating-axis or GJK/EPA checks, and precomputed structures (no-fit relations in 2D; extreme points or empty maximal spaces in 3D). Hybrid designs are common: geometry for candidate generation and feasibility, rasterized descriptors for local voids/support [ARCO22].

Candidate placements (“interesting points”). Modern 3D solvers avoid global pose search by enumerating high-quality placements at the evolving packing frontier. Two families dominate: (i) *extreme points*, induced by faces/edges/vertices of the container and already placed items, and (ii) *empty maximal spaces*, maintained as items are inserted. Both admit quick filters for container bounds, non-overlap, minimal support, separations, and admissible orientations [ARCO22].

Neural scoring and learned proposals. Neural components most often re-rank EP/EMS candidates instead of proposing arbitrary poses. Feature sets typically mix void/waste metrics, clearances, support/contact estimates, and simple surrogates for load distribution or access. Lightweight models (MLPs, attention over placed items) can replace hand-crafted tie-breakers without changing the surrounding heuristic or matheuristic scaffold [ARCO22].

Constructive heuristics. Constructive backbones couple item ordering (e.g., by size, shape complexity, or “hardness”) with candidate evaluation rules (minimize residual void, maximize support, minimize overhang, respect separations and orientation sets). These rules deliver fast feasible layouts and provide consistent starting points for improvement phases [ARCO22].

Local improvement. After a feasible placement, local refinements (swap, rotate within allowed sets, reinsert, neighborhood re-pack) and compaction steps reduce fragmentation. Physics-inspired or projection-based moves gently compress the layout; when overlaps appear, constraints are restored by expanding or re-projecting to the feasible set. Such micro-optimizations target the highest-void regions and are effective under strict time budgets [ARCO22].

Metaheuristics and matheuristics. GRASP, VNS, tabu, and LNS frameworks explore item orderings and partial re-packs; matheuristics embed exact subproblems (e.g., orientation/placement pricing, stability/separation checks) to sharpen feasibility and improve bounds. This hybridization remains the most common route to high fill ratios with realistic industrial constraints [ARCO22].

2D, 3D online, offline rasterisation, the one where they all bunch up and expand convex concave integration into different problems interesting points, using NN to find them different heuristics used

3.2 Evolutionary Neural Architecture Search

Neurogenesis, ENAS different proposed layers, functions description of representations of nn used in literature graph of architecture neural structure vs hyperparameters, which ones

3.3 Traveling Salesman Problem

different solvers deterministic vs nondeterministic, gauge speed NN to solve
in combination with bin packing



Chapter 4

Proposed method

stuff from research, make some graphs, callback to problem definition in ch2



Chapter 5

Implementation



Chapter 6

Experiments



Chapter 7

Conclusions

Appendix A

Bibliography

- [AMSAVP22] M.T. Alonso, A. Martinez-Sykora, R. Alvarez-Valdes, and F. Parreño, *The pallet-loading vehicle routing problem with stability constraints*, European Journal of Operational Research **302** (2022), no. 3, 860–873.
- [ARCO22] Sara Ali, António Galvão Ramos, Maria Antónia Carravilla, and José Fernando Oliveira, *On-line three-dimensional packing problems: A review of off-line and on-line solution approaches*, Computers & Industrial Engineering **168** (2022), 108122.
- [BB24] Reinhard Booysen and Anna Sergeevna Bosman, *Multi-objective evolutionary neural architecture search for recurrent neural networks*, Neural Processing Letters **56** (2024), no. 4.
- [BTC23] Guillem Bonet Filella, Alessio Trivella, and Francesco Corman, *Modeling soft unloading constraints in the multi-drop container loading problem*, European Journal of Operational Research **308** (2023), no. 1, 336–352.
- [dNAJ21] Oliviana Xavier do Nascimento, Thiago Alves de Queiroz, and Leonardo Junqueira, *Practical constraints in the container loading problem: Comprehensive formulations and exact algorithm*, Computers & Operations Research **128** (2021), 105186.
- [EMH19] Thomas Elsken, Jan Hendrik Metzen, and Frank Hutter, *Neural architecture search: A survey*, Journal of Machine Learning Research **20** (2019), no. 55, 1–21, Editor: Sebastian Nowozin.
- [GOGL16] A. Galvão Ramos, José F. Oliveira, José F. Gonçalves, and Manuel P. Lopes, *A container loading algorithm with static*

- mechanical equilibrium stability constraints*, Transportation Research Part B: Methodological **91** (2016), 565–581.
- [GPAAVP23] I. Gimenez-Palacios, M.T. Alonso, R. Alvarez-Valdes, and F. Parreño, *Multi-container loading problems with multidrop and split delivery conditions*, Computers & Industrial Engineering **175** (2023), 108844.
- [GTMP22] Mikele Gajda, Alessio Trivella, Renata Mansini, and David Pisinger, *An optimization approach for a complex real-life container loading problem*, Omega **107** (2022), 102559.
- [KEK21] Corinna Krebs, Jan Fabian Ehmke, and Henriette Koch, *Advanced loading constraints for 3d vehicle routing problems*, OR Spectrum **43** (2021), no. 4, 835–875.
- [LSX⁺20] Yuqiao Liu, Yanan Sun, Bing Xue, Mengjie Zhang, and Gary G. Yen, *A survey on evolutionary neural architecture search*, CoRR **abs/2008.10937** (2020).
- [RMS⁺17] Esteban Real, Sherry Moore, Andrew Selle, Saurabh Saxena, Yutaka Leon Suematsu, Jie Tan, Quoc Le, and Alex Kurakin, *Large-scale evolution of image classifiers*, 2017.
- [TK01] M. Takahashi and H. Kita, *A crossover operator using independent component analysis for real-coded genetic algorithms*, Proceedings of the 2001 Congress on Evolutionary Computation (IEEE Cat. No.01TH8546), vol. 1, 2001, pp. 643–649 vol. 1.
- [WHS07] Gerhard Wäscher, Heike Haußner, and Holger Schumann, *An improved typology of cutting and packing problems*, European Journal of Operational Research **183** (2007), no. 3, 1109–1130.
- [WSS⁺23] Colin White, Mahmoud Safari, Rhea Sukthanker, Binxin Ru, Thomas Elsken, Arber Zela, Debadepta Dey, and Frank Hutter, *Neural architecture search: Insights from 1000 papers*, 2023.

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ZADÁNÍ BAKALÁŘSKÉ PRÁCE

Pro: Tomáš Hejda
Obor: Matematické inženýrství
Zaměření: Matematické modelování
Název práce: Sprátelené morfismy na sturmovských slovech / Amicable Morphisms on Sturmian Words

Osnova:

1. Seznamte se se základními pojmy a větami z teorie symbolických dynamických systémů.
2. Udělejte rešerši poznatků o sturmovských slovech: přehled ekvivalentních definic sturmovských slov, popis morfismů zachovávajících sturmovská slova, popis standardních párů slov.
3. Zkoumejte vlastnosti párů sprátelených sturmovských morfismů, pokuste se popsat jejich generování a počty v závislosti na tvaru jejich matice.

Doporučená literatura:

1. M. Lothair, Algebraic Combinatorics on Words, Encyclopedia of Math. and its Applic., Cambridge University Press, 1990
2. J. Berstel, Sturmian and episturmian words (a survey of some recent result results), in: S. Bozapalidis, G. Rahonis (eds), Conference on Algebraic Informatics, Thessaloniki, Lecture Notes Comput. Sci. 4728 (2007), 23-47.
3. P. Ambrož, Z. Masáková, E. Pelantová, Morphisms fixing a 3iet words, preprint DI (2008)

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